STUDENT PROJECTS LABORATORY - LOCKOUT / TAGOUT [LOTO] PROCEDURE

ENERGY ISOLATION SPECIFICATION | ASSET: WLD-01 | TAG: TIG-275-01

Lincoln Precision TIG 275 Welder #1

Authorized LOTO Personnel: Sr. Instructional Lab Supervisor (Max Wright) & Student Research Assistants (SRAs)

Notification Channel: Max Wright - Teams / Text Message | Zone: Zone 3: Welding & Thermal Processing (WLD)

STEP 1 - NORMAL MACHINE SHUTDOWN PROCEDURE

Extinguish arc, allow post-flow timer to cool tungsten, toggle main power switch OFF (Front Panel Upper Right).

STEP 2 - HAZARDOUS ENERGY SOURCES, MAGNITUDES & ISOLATION POINTS

Source #1: Electrical (240V AC (High-Capacitance Inverter/Transformer))

Isolation Location: Wall switch box disconnect behind welder.

Lockout Procedure: Pull wall switch box lever down to OFF & install LOTO lock; WAIT AT LEAST 5 MINUTES for internal capacitors to dissipate.

Source #2: Pneumatic (Shielding Gas) (2,500 PSI Compressed Argon Cylinder)

Isolation Location: Argon cylinder top valve & regulator assembly.

Lockout Procedure: Turn cylinder top valve clockwise until all yellow (no red showing), remove gas line from regulator, depress foot pedal to purge solenoids/lines, and verify both regulator gauges read 0 PSI.

STEP 3 - STORED / RESIDUAL ENERGY DISSIPATION

Wait >= 5 minutes for capacitor bleed; depress foot pedal with gas disconnected to purge trapped Argon pressure.

STEP 4 - ZERO-ENERGY VERIFICATION ('TRY' STEP)

Verify display is dark and cooling fan does not engage when power switch is toggled; confirm both regulator gauges read 0 PSI.